

Digital Modulation

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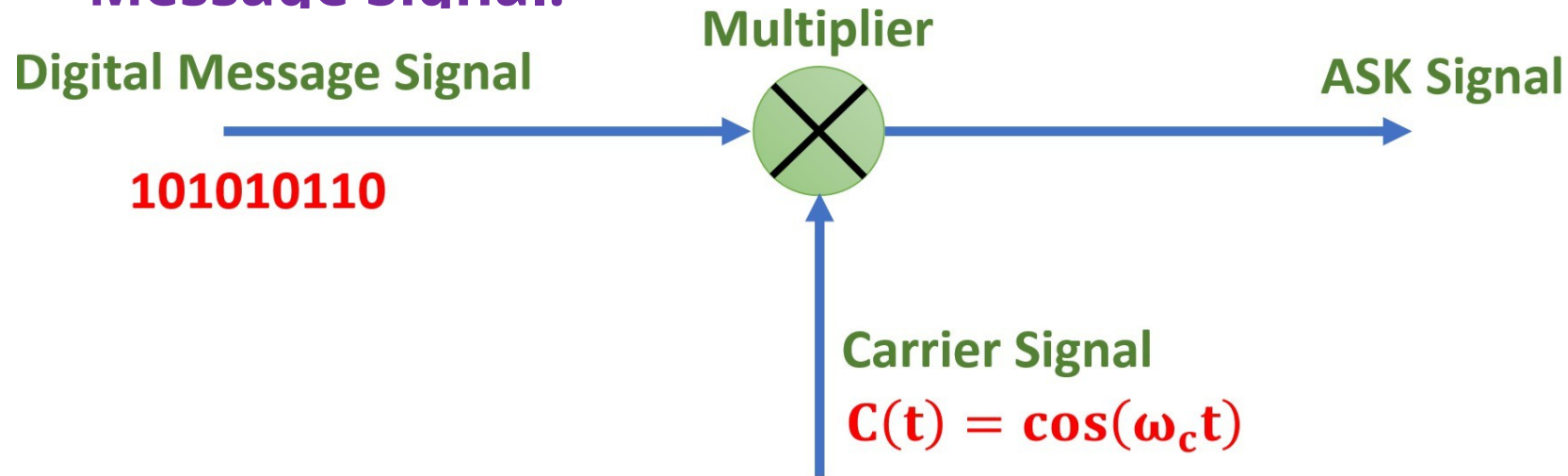
Amplitude Shift Keying -

Outline

- ❑ Basics of ASK
- ❑ Waveforms of ASK
- ❑ Bandwidth of ASK
- ❑ Demodulation of ASK
- ❑ Constellation Diagram of ASK
- ❑ Advantages of ASK
- ❑ Disadvantages of ASK
- ❑ Applications of ASK

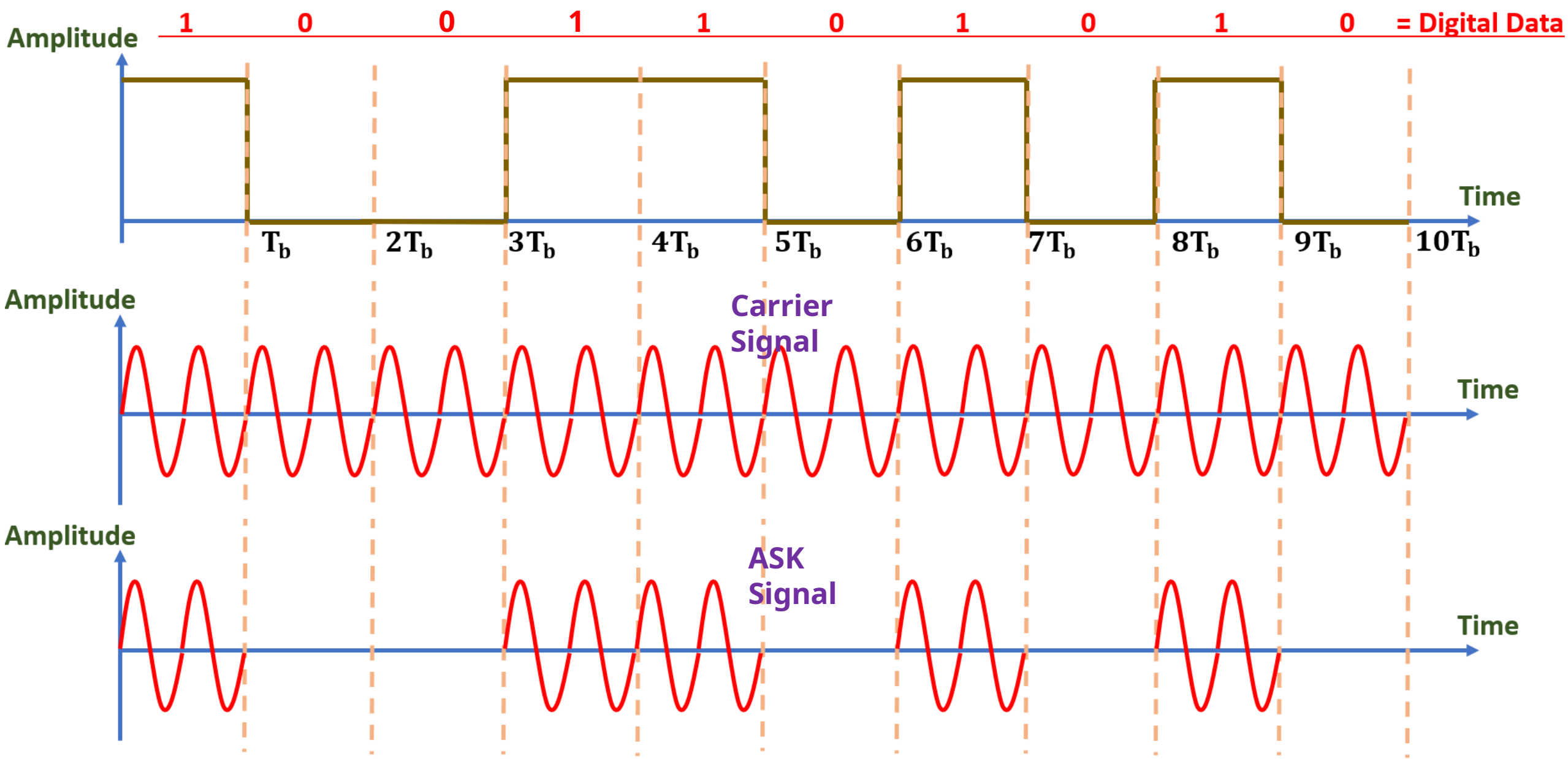
Basics of ASK

- ASK is a digital modulation technique that converts digital signals into Analog form by varying the carrier's amplitude, enabling efficient transmission over a communication channel.
- In ASK, the Amplitude of the Carrier Signal varies according to the Digital Message Signal.

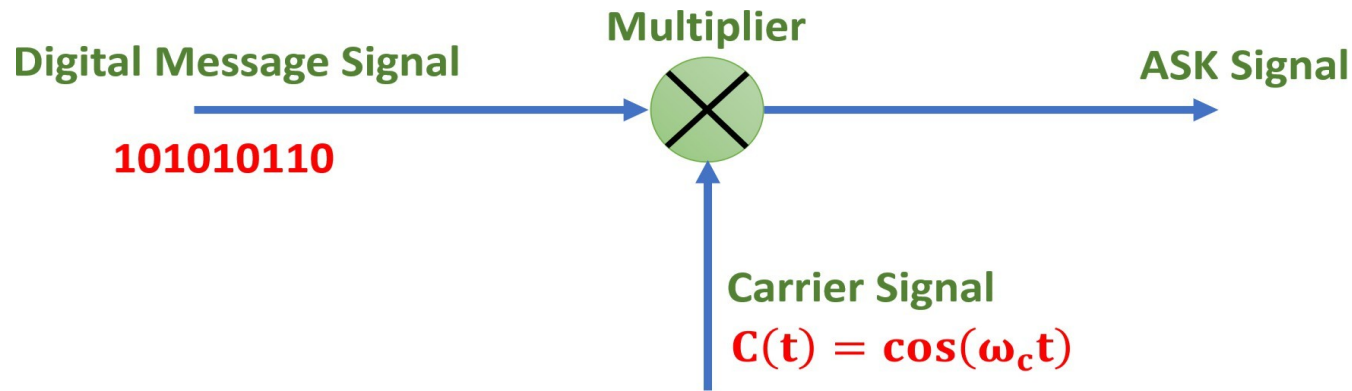


- {BASK or OOK} ASK Signal is given by:
 $\Rightarrow \phi_{ASK} = C(t)$ For Digital data 1
 $\phantom{\Rightarrow \phi_{ASK}} = 0$ For Digital data 0

Waveforms of ASK



Bandwidth of ASK



- Data Rate (Bit Rate, R)
👉 Number of bits transmitted per second (bps)
- Baud Rate (Symbol Rate, S)
👉 Number of signal changes (symbols) per second (baud)

🔗 Relationship

$$R = S \times \log_2(M)$$

👉 Where:

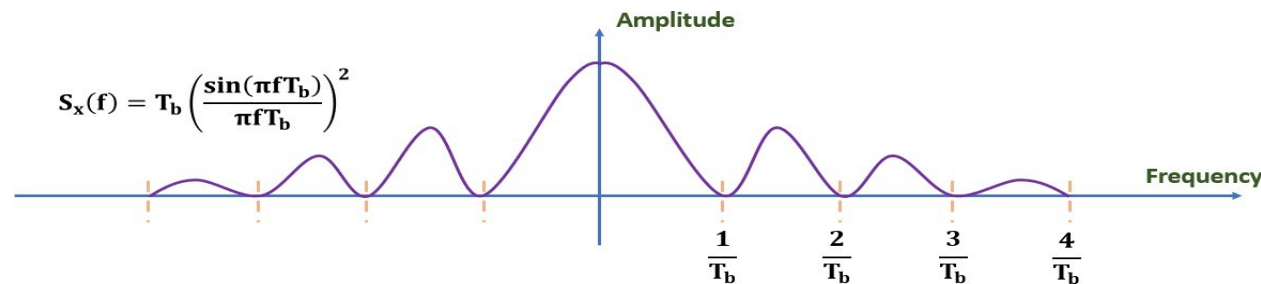
- M = number of signal levels (constellation points)

❑ Ideally, its bandwidth equals the Baud rate of the message, but in practice, it is slightly

greater

$$\Rightarrow BW_{ASK} = (1 + \alpha)r$$

- r is Baud Rate of Message Signal
- α is Roll off Factor depends on Filtering



$$\Rightarrow BW_{min} = 2F_b$$

- At Nyquist Rate $F_b = R/2$ (For Binary ASK)
- R is Data rate.
- F_b = Highest baseband frequency
- For M-ary ASK, $F_b = \frac{\text{Baud Rate}}{2}$

Another Way: (For M-ary ASK)

When α is zero, BW will be minimum. So,

- $BW_{min} = (1 + 0)r = r \Rightarrow \text{Baud rate}$

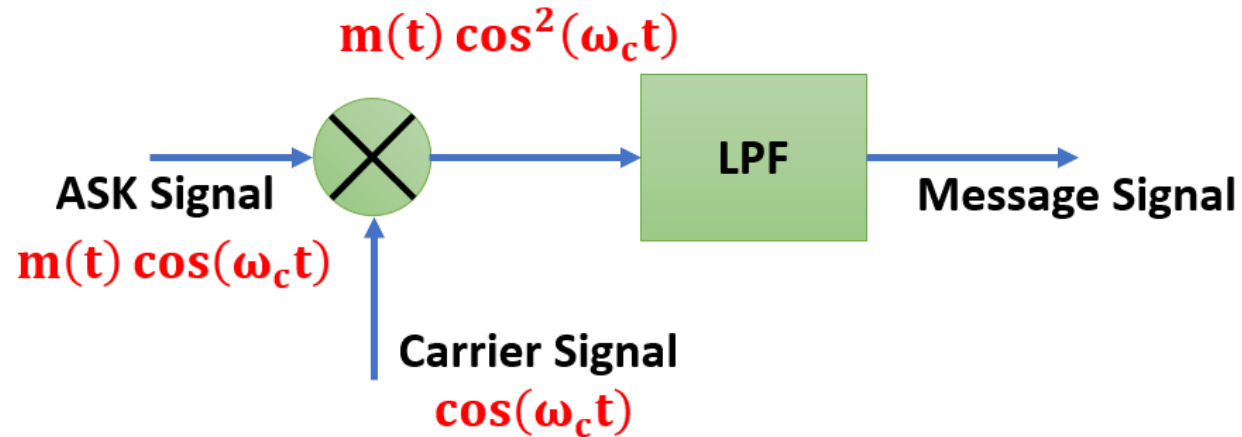
Demodulation of ASK

□ There are two Types of ASK

Demodulation:

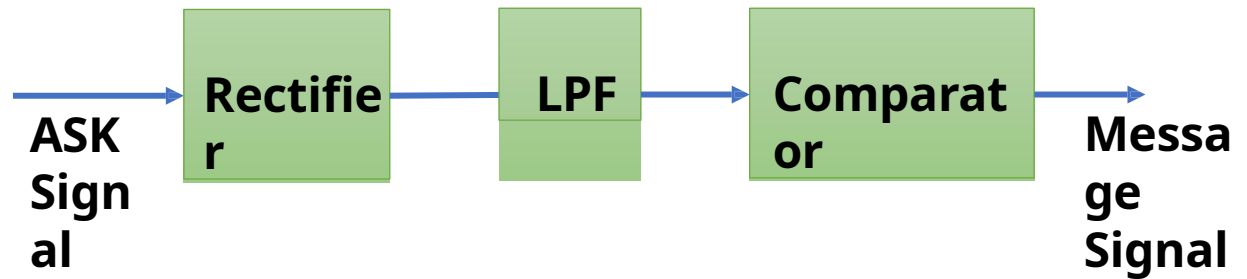
- Synchronous (Coherent)
- Asynchronous (Non-Coherent)

Synchronous (Coherent) ASK Demodulation



- ✓ Pro: Efficient Method.
- ✓ Con: Need of Oscillator for Carrier.

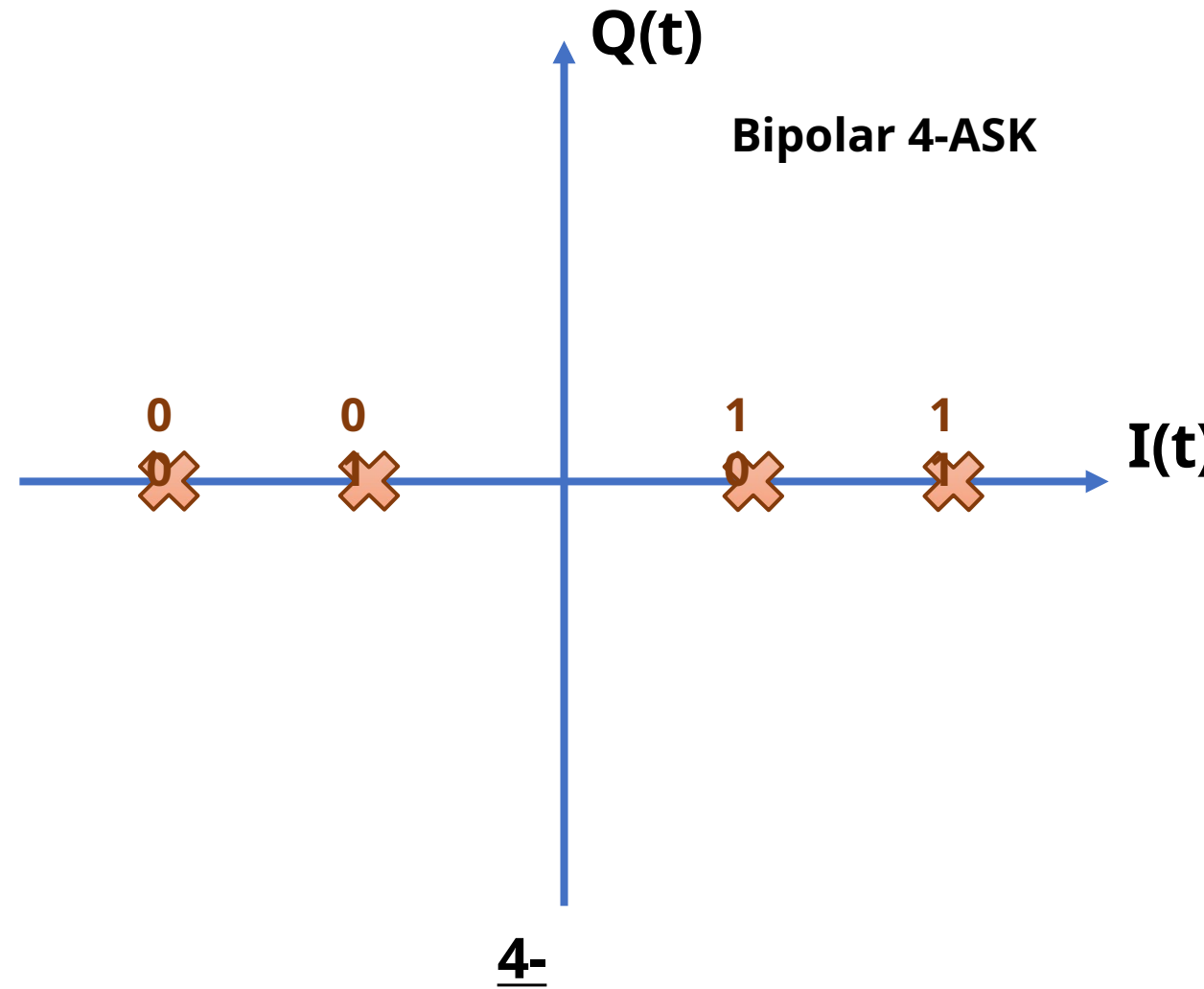
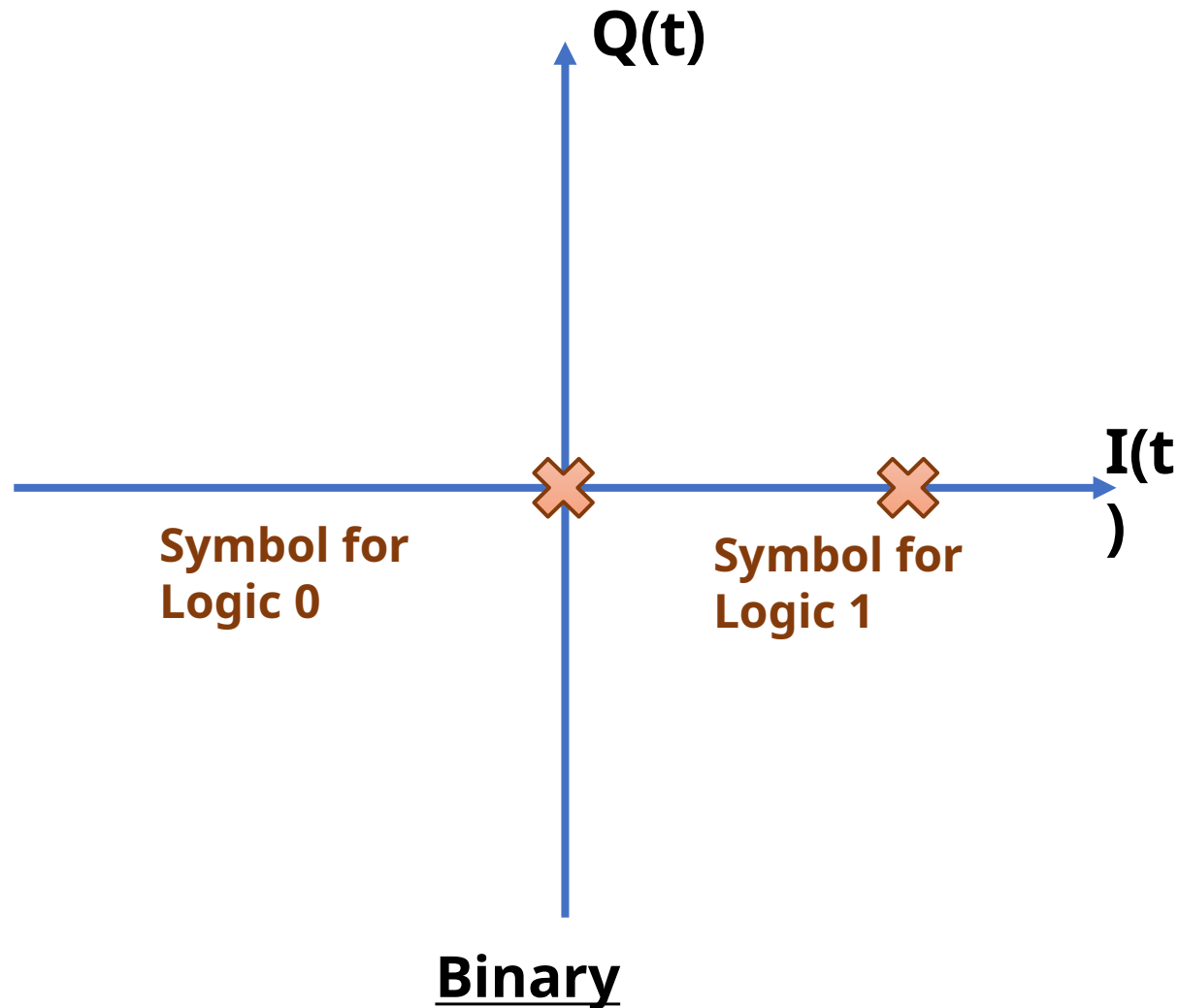
Asynchronous (Non-Coherent) ASK Demodulation



- ✓ Pro: No Need of Oscillator for Carrier.
- ✓ Con: Poor Performance (Less SNR)

Constellation Diagram of ASK

- A constellation diagram is a graphical representation of a digital signal showing how data symbols are mapped onto signal states. $I(t) \rightarrow$ In-phase component and $Q(t) \rightarrow$ Quadrature component (90° phase shifted)



Advantages of ASK

- ❑ Simple Implementation (Easy Modulation and Demodulation)
- ❑ Band Efficient
- ❑ Useful for Low Cost Applications

Disadvantages of ASK

- ❑ Highly Sensitive to Noise
- ❑ Bad Performance in Wireless Communication

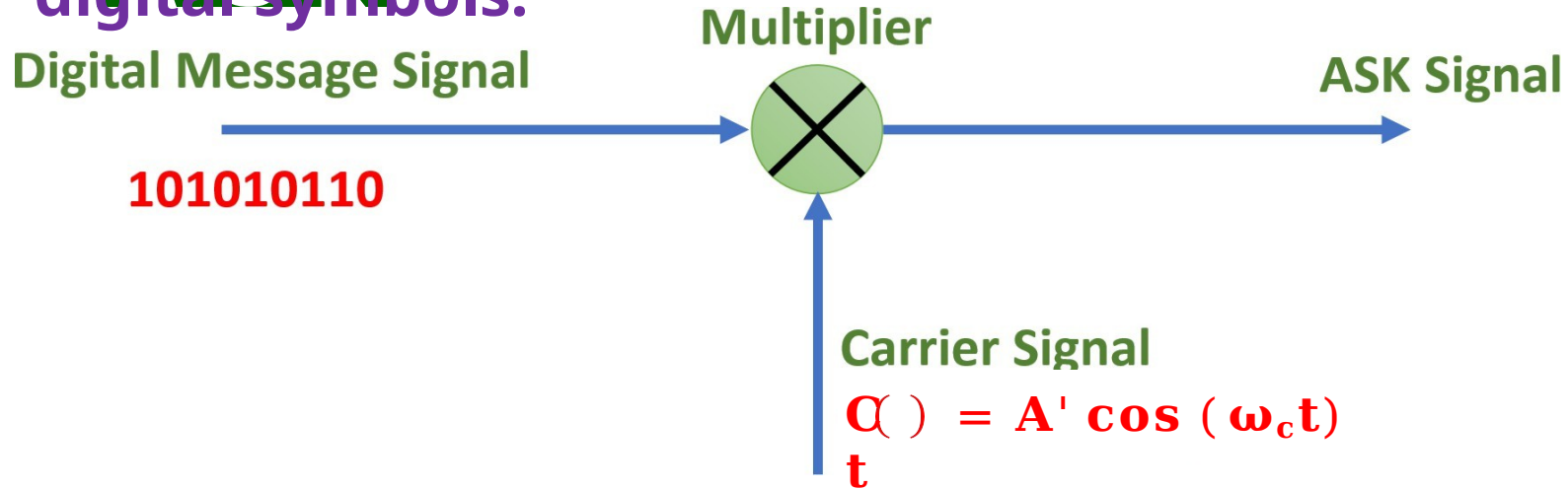
Applications of ASK

- ❑ Optical Fiber Communication
- ❑ Low Frequency RF Communication {RFID Tag}
- ❑ In Early Modems
- ❑ IR Remote Control

Signal Space Diagram of

ASK

- Signal Space Diagram explains the distance between digital symbols.



⇒ Energy of ASK Signal.

$$\begin{aligned}\Rightarrow E &= \int_0^{T_b} \phi_{ASK}^2 dt \\ &= \int_0^{T_b} A^2 \cos^2(\omega_c t) dt \\ &= \frac{A^2}{2} \int_0^{T_b} (1 + \cos(2\omega_c t)) dt\end{aligned}$$

$$\Rightarrow E = \frac{A^2 T_b}{2}$$

$$\Rightarrow A = \sqrt{\frac{2E}{T_b}}$$

$$\Rightarrow E = \int_0^{T_b} A^2 \cos^2 \omega_c t dt$$

$$\Rightarrow E = \int_0^{T_b} \left(\sqrt{\frac{2E}{T_b}}\right)^2 \cos^2 \omega_c t dt$$

⇒ Power of ASK Signal

$$\Rightarrow P_s = \frac{1}{T_b} \int_0^{T_b} \phi_{ASK}^2 dt$$

$$\Rightarrow P_s = A^2/2$$

$$\Rightarrow A = \sqrt{2P_s}$$

$$\Rightarrow 1 = \int_0^{T_b} \left(\sqrt{\frac{2}{T_b}}\right)^2 \cos^2 \omega_c t dt$$

⇒ If we take amplitude of carrier $\sqrt{\frac{2}{T_b}}$ then energy of carrier will be unity.

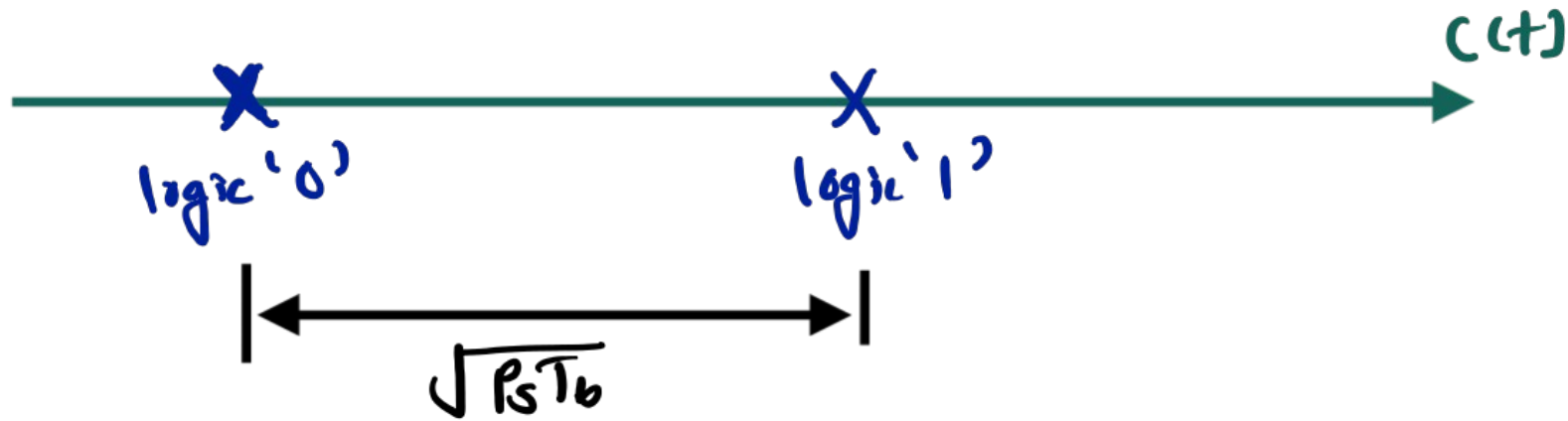
$$\Rightarrow A' = \sqrt{\frac{2}{T_b}}$$

⇒ ASK signal

$$\Rightarrow \phi_{ASK} = \sqrt{2P_s} \cos \omega_c t \quad \text{— For logic '1'}$$

$$= \sqrt{P_s T_b} \sqrt{\frac{2}{T_b}} \cos \omega_c t$$

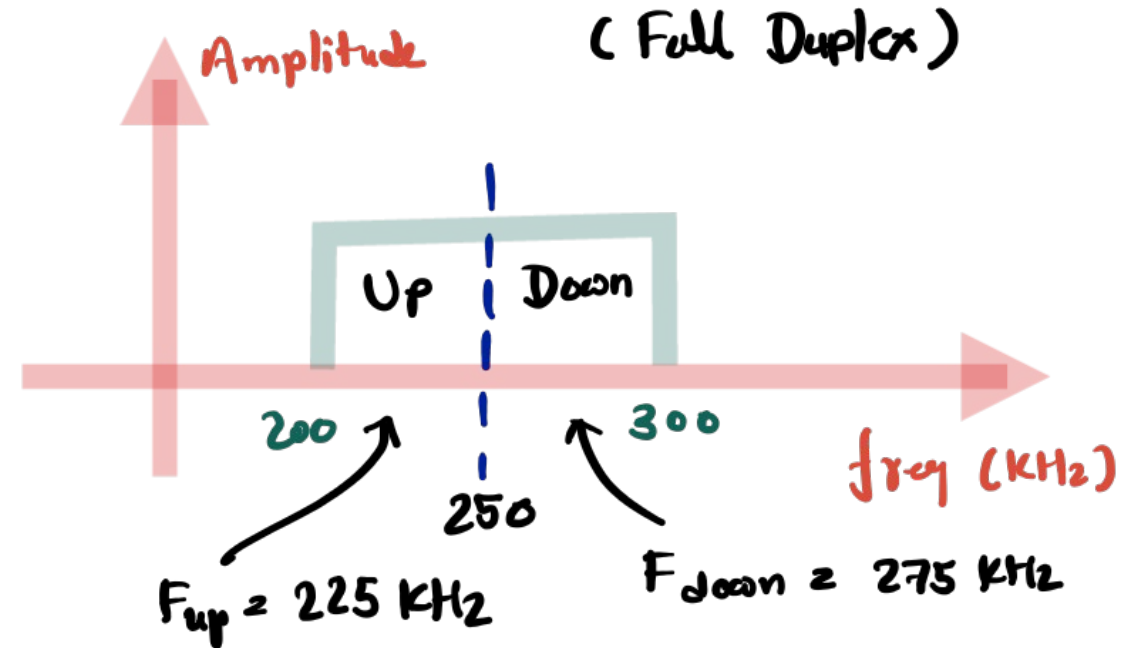
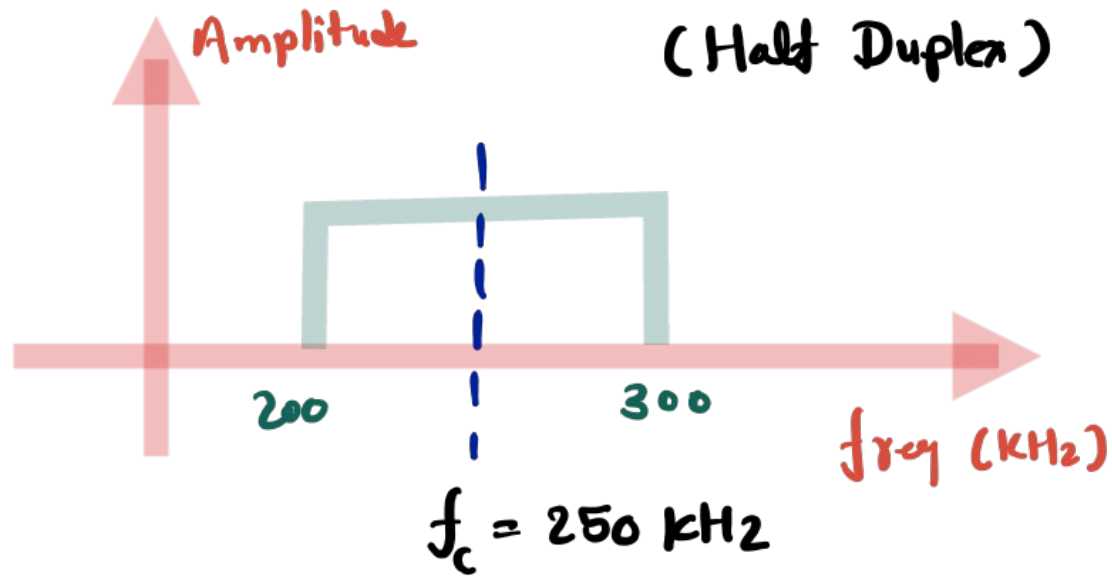
$$= \sqrt{P_s T_b} \quad c(t)$$



Example of

ASK

1. **Example:** We have an available bandwidth of 100 kHz, ranging from 200 kHz to 300 kHz. If we modulate our data using BASK with a roll-off factor of 1, what are the carrier frequency and bit rate?



$\Rightarrow BW = 100 \text{ kHz}$

$n = 1$

$\alpha = 1$

$$\leadsto B\omega = (1+2)\gamma$$

$$\Rightarrow \gamma = B\omega / (1+2)$$

$$= \frac{100}{2}$$

$$= 50 \text{ kband}$$

$$\leadsto R = \gamma n$$

$$= 50 \times 1$$

$$= 50 \text{ Kbps}$$

Frequency Shift Keying -

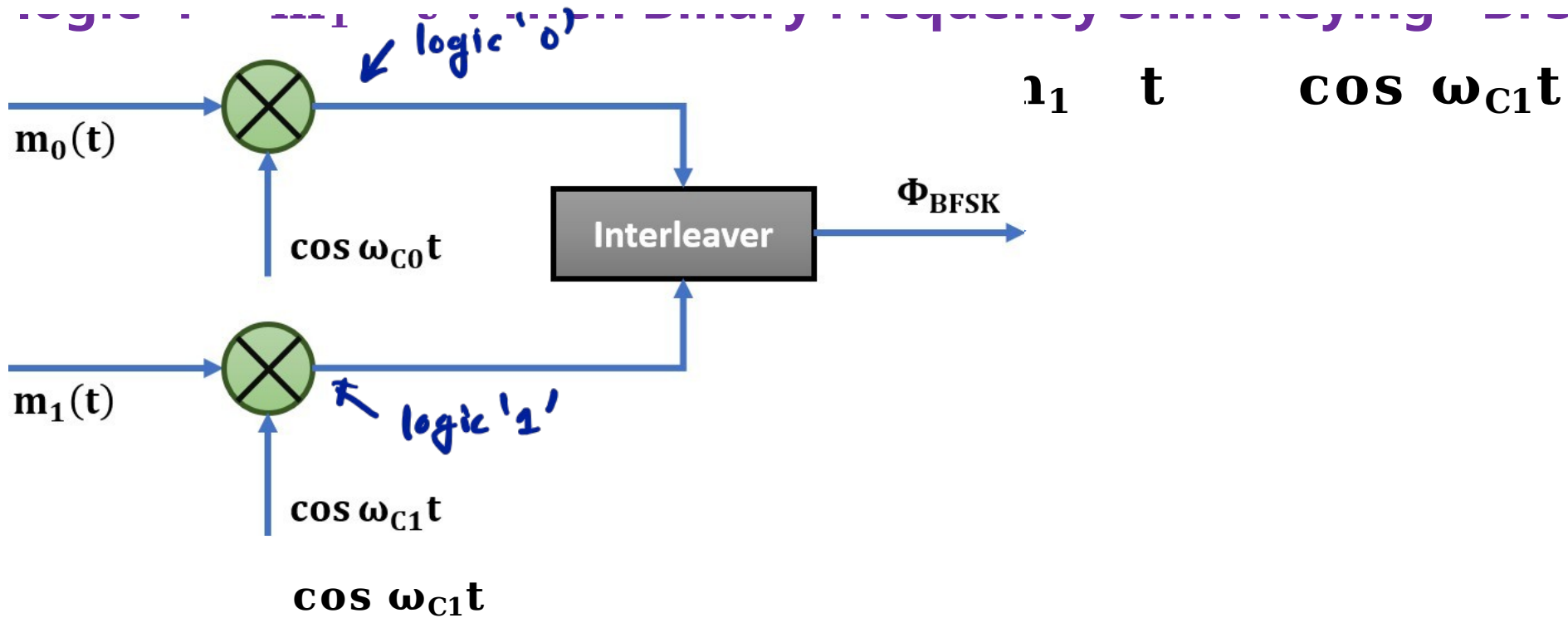
Outline

- ☐ Basics of FSK
- ☐ Waveforms of FSK
- ☐ Bandwidth of FSK
- ☐ Multilevel FSK
- ☐ Demodulation of FSK
- ☐ Constellation Diagram of FSK
- ☐ Advantages of FSK
- ☐ Disadvantages of FSK
- ☐ Applications of FSK

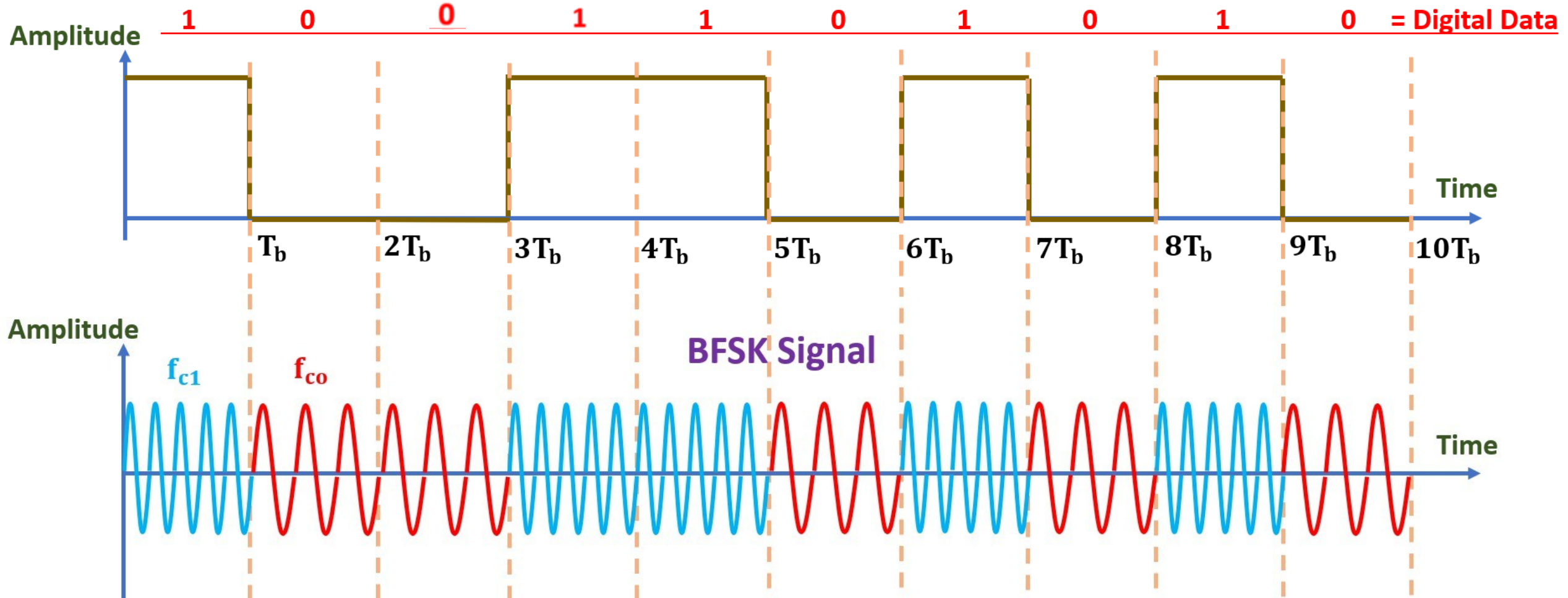
Basics of FSK

- FSK is a digital modulation technique that converts digital signals into Analog form by varying the carrier's frequency, enabling efficient transmission over a communication channel.
- In FSK, the Frequency of the Carrier Signal varies according to the Digital
- Let's consider the message Signal $m(t)$, which is a combination of logic '0' = $m_0(t)$ and logic '1' = $m_1(t)$. Then Binary Frequency Shift Keying - BFSK is given by:

$$\Rightarrow \Phi_{\text{BFSK}} = m_0(t) \cos[\omega_{c0}t] + m_1(t) \cos[\omega_{c1}t]$$



Waveforms of FSK



BFSK Signal Basics

In binary FSK (BFSK), two frequencies are used:

$$s(t) = \begin{cases} A \cos(2\pi f_1 t), & \text{bit 1} \\ A \cos(2\pi f_2 t), & \text{bit 0} \end{cases}$$

Define:

- Center frequency:

$$f_c = \frac{f_1 + f_2}{2}$$

- Frequency deviation:

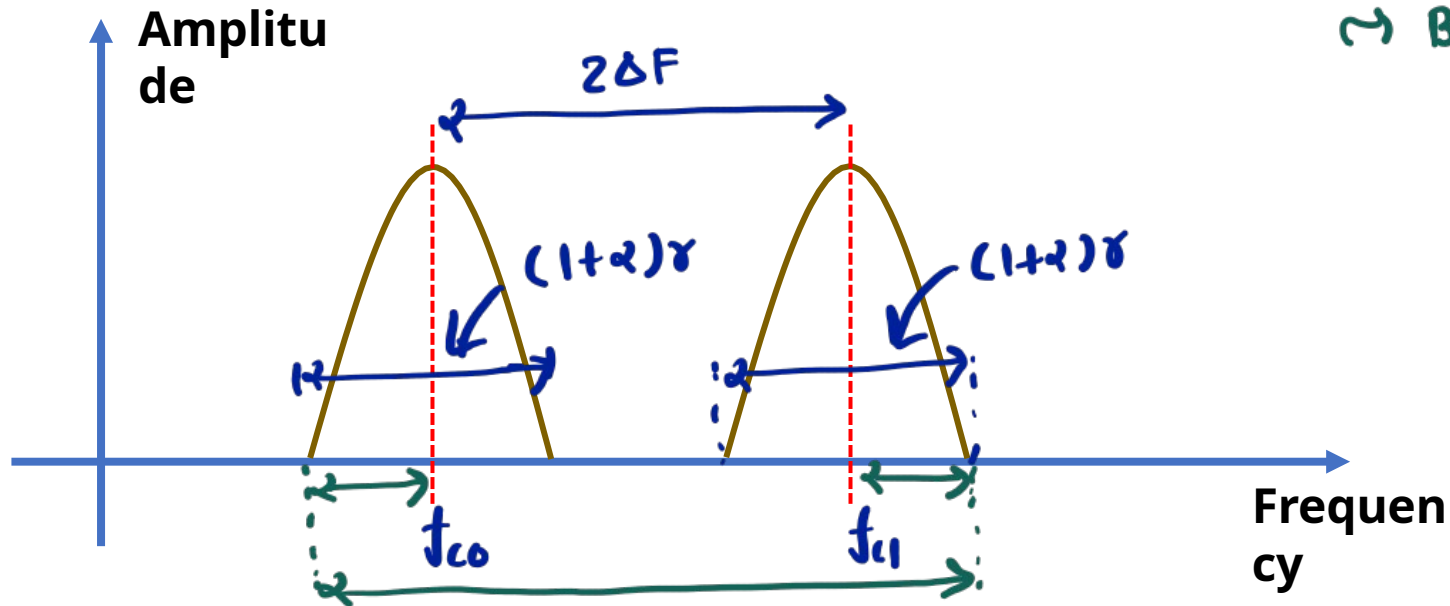
$$\Delta f = \frac{f_1 - f_2}{2}$$

So:

$$f_1 = f_c + \Delta f, \quad f_2 = f_c - \Delta f$$

$2\Delta f = \text{Frequency Separation}$

Bandwidth of BFSK



$$\begin{aligned} \Rightarrow BW &= 2\Delta F + \frac{(1+\alpha)B}{2} + \frac{(1+\alpha)B}{2} \\ &= (1+\alpha)B + 2\Delta F \end{aligned}$$

□ Minimum Bandwidth of

$$(\alpha = 0)$$

$$\begin{aligned} BW_{\min} &= 2(1+\alpha)B \\ &= 2B \end{aligned}$$

Multi-Level FSK

Number of Level=Number of Symbol =M

~) For BFSK

$$- M = 2 = 2^1 = 2^n \leadsto n=1 \leadsto f_{c0}, f_{c1}$$

~) For 4-FSK

$$- M = 4 = 2^2 = 2^n \leadsto n=2 \leadsto f_{c0}, f_{c1}, f_{c2}, f_{c3}$$

~) For 8-FSK

$$- M = 8 = 2^3 = 2^n \leadsto n=3 \leadsto f_{c0}, f_{c1}, \dots, f_{c7}$$

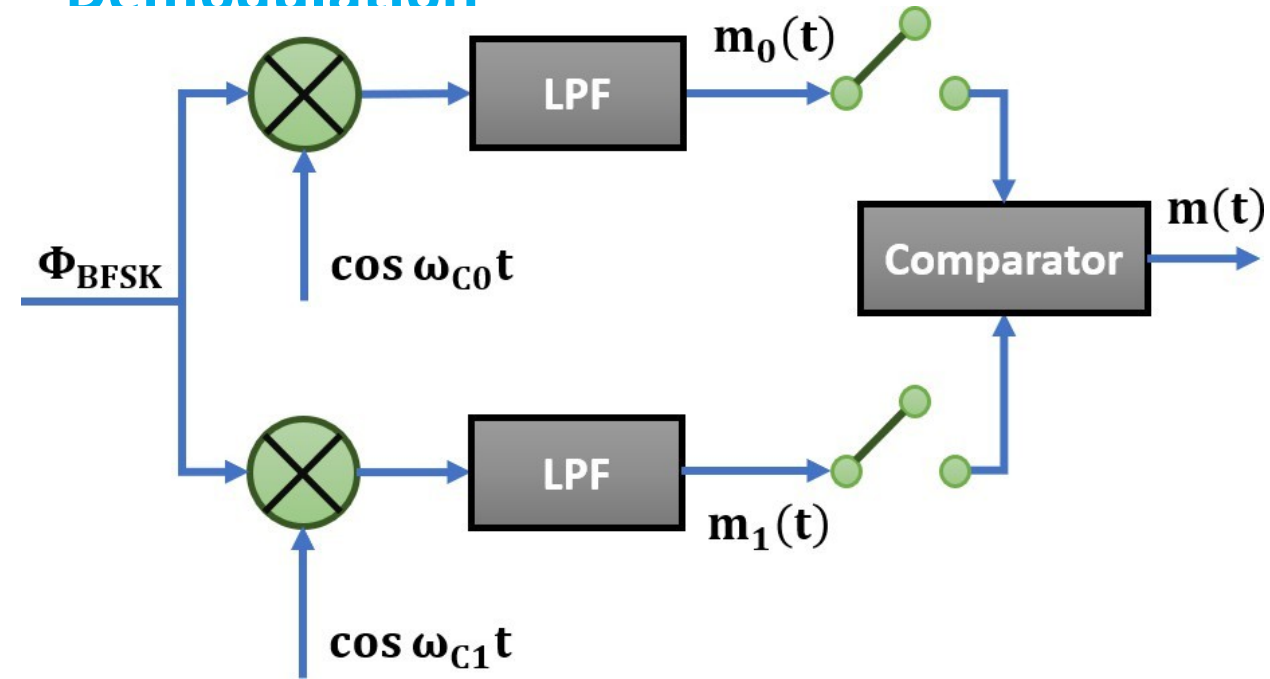
~) BW of M levels FSK

$$\Rightarrow BW = (1+2)\gamma + 2\Delta F(M-1)$$

$$\begin{aligned} \Rightarrow BW_{\min} &= (1+0)\gamma + (1+0)\gamma(M-1) \\ &= \gamma M \end{aligned}$$

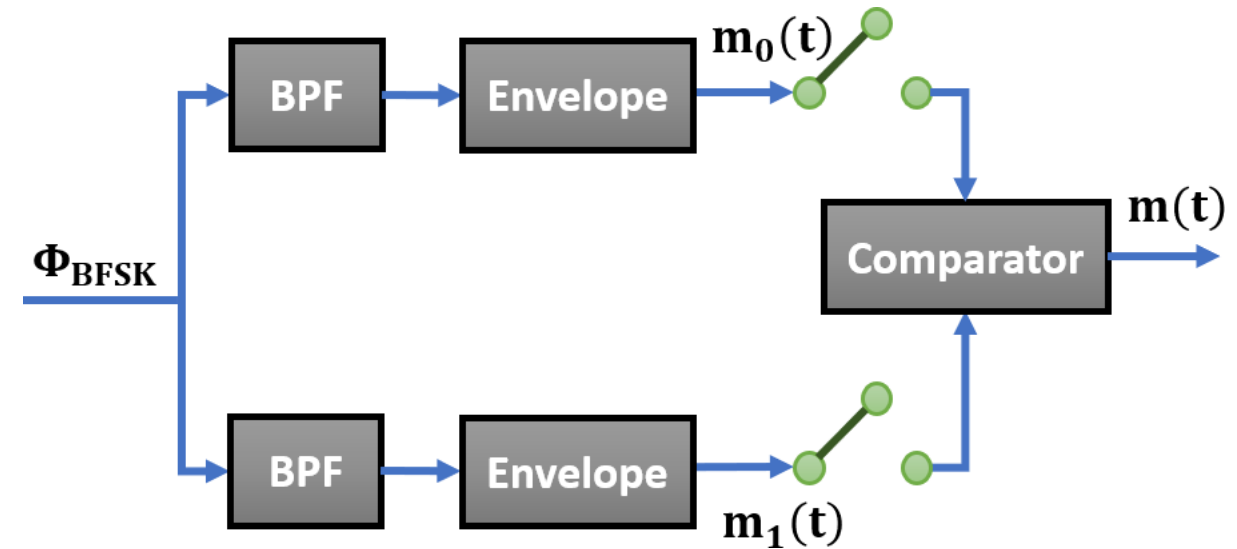
FSK Demodulation

Synchronous (Coherent) FSK Demodulation



- ✓ Pro: Efficient Method.
- ✓ Con: Need of Oscillator for Carrier.

Asynchronous (Non-Coherent) FSK Demodulation



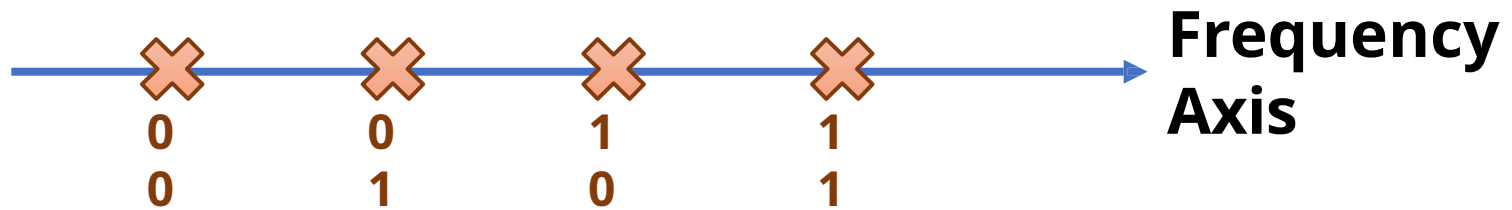
- ✓ Pro: No Need of Oscillator for Carrier.
- ✓ Con: Poor Performance (Less SNR)

Constellation Diagram of FSK

❑ For BFSK



❑ For 4-FSK



Advantages of FSK

- ❑ Simple Implementation (Easy Modulation and Demodulation)
- ❑ Better Noise Immunity compared to ASK
- ❑ Reliable for low Data Rate Applications

Disadvantages of FSK

- ❑ Required Large Bandwidth compared to ASK and PSK
- ❑ Coherent Detection more complex
- ❑ Not good for high Data Rate Transmission

Applications of FSK

- ❑ In Early Modems
- ❑ Low Frequency Wireless Systems
- ❑ Bluetooth (GFSK)
- ❑ Paging Systems

Examples of

1. **Example:** We transmit data 3 bits at a time at a bit rate of 3 Mbps. The carrier frequency is 10 MHz. Calculate (a) the number of levels, (b) the baud rate, and (c) the minimum bandwidth for FSK.

$$\leadsto n = 3$$

$$R = 3 \text{ Mbps}$$

$$f_c = 10 \text{ MHz}$$

$$\leadsto M = 2^n = 2^3 = 8$$

$$\leadsto \gamma = \frac{R}{n} = \frac{3}{3} = 1 \text{ Mbaud}$$

$$\leadsto BW_{\min} = \gamma M \\ = 8 \text{ MHz}$$

2. Example: We have an available bandwidth of 100 kHz, spanning from 200 kHz to 300 kHz. If we modulate our data using binary FSK with a roll-off factor 1 and frequency deviation is 25kHz, what should be the carrier frequency and the bit rate?

Solution:

For Binary FSK, $n=1$

Roll-off factor, $\alpha = 1$

BW=100kHz

Frequency Deviation, $\Delta F = 25\text{kHz}$

For Binary FSK,

$$\text{BW} = 2\Delta F + (1 + \alpha)r \text{ [Here, } r \text{ is Baud Rate]}$$

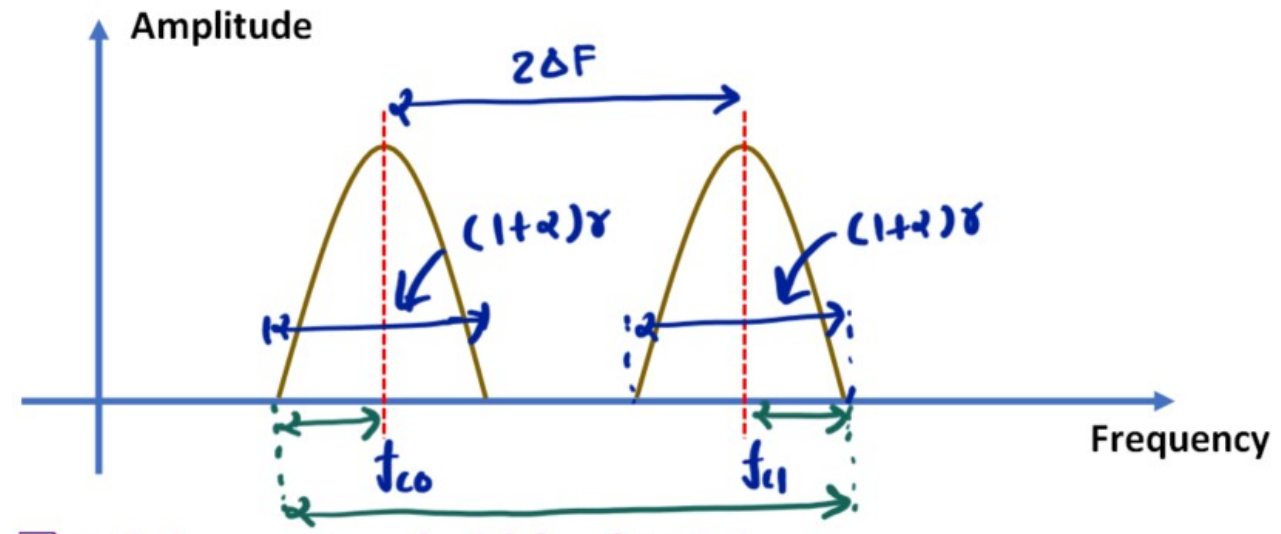
$$\Rightarrow 100 = 2 \times 25 + (1 + 1) \times r$$

$$\Rightarrow r = 25 \text{ kbaud}$$

so, Bit Rate, $R=nr=1*25 \text{ kbps}=25 \text{ kbps}$

$$\text{Carrier Frequency, } f_{c0} = 200 + \frac{(1+\alpha)r}{2} = 200 + 25 = 225 \text{ kHz}$$

$$\text{Carrier Frequency, } f_{c1} = f_{c0} + 2\Delta F = 225 + 50 = 275 \text{ kHz}$$



M-ary FSK - MFSK

Basics and Equation of MFSK

⇒ m = Modulation Order or Number of Symbols

$$\Rightarrow m = 2^n \Rightarrow n = \log_2 m$$

↑
Number of bits per Symbol

⇒ For BFSK

$$x_1(t) = A \cos(2\pi f_1 t) \quad \text{— For logic '0'}$$

$$x_2(t) = A \cos(2\pi f_2 t) \quad \text{— For logic '1'}$$

⇒ For MFSK

$$x_i(t) = A \cos(2\pi f_i t) = \sqrt{2P_s} \cos(2\pi f_i t)$$

where, $i = 1, 2, 3, \dots, m$

Symbol Duration of MFSK

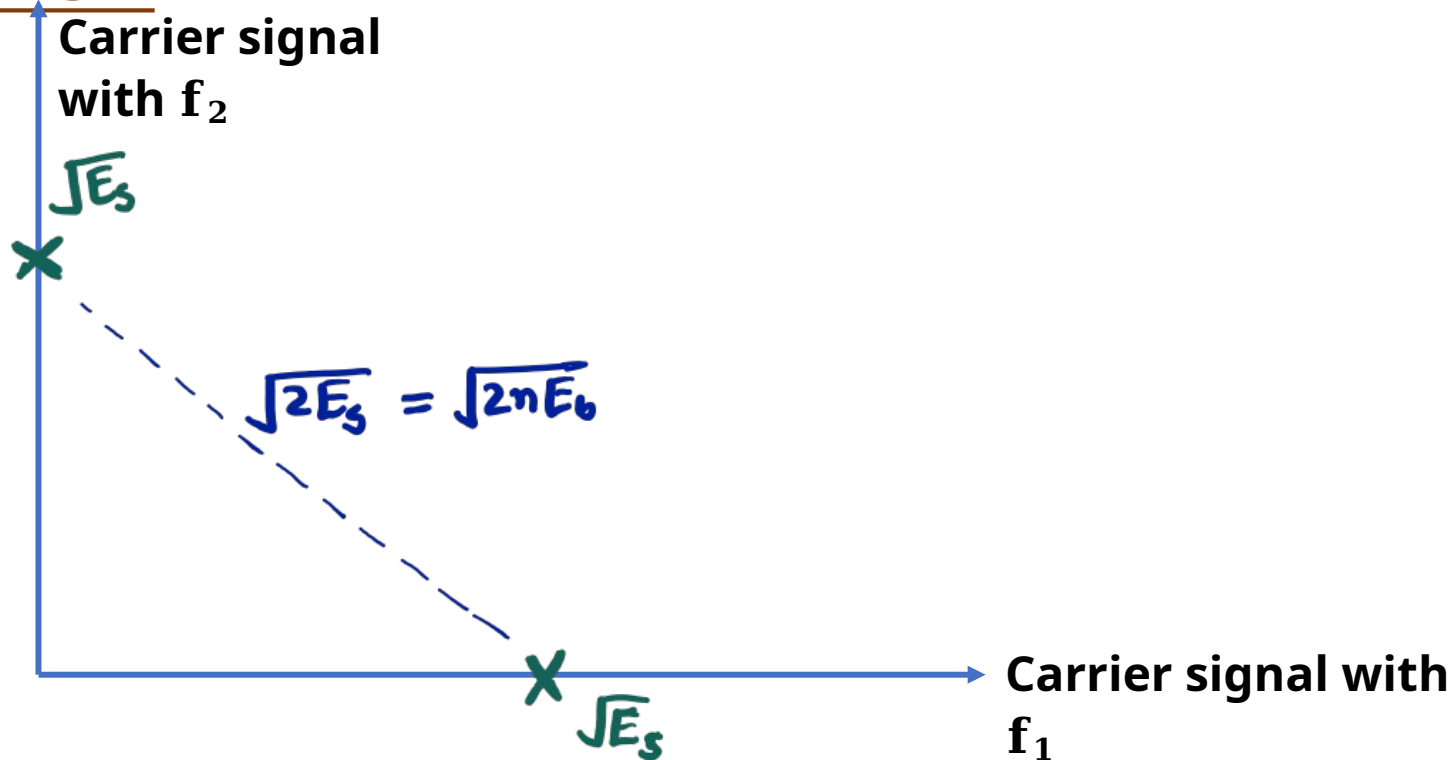
$$\Rightarrow T_s = n T_b$$

T_s = Symbol Duration

T_b = Bit Duration

n = number of bits per symbol

Spacing between symbols in MFSK



Frequencies of MFSK

→ There are M number of Carrier Frequencies in MFSK.

→ i^{th} Frequency is given by

$$f_i = f_c + (2i - M - 1)\Delta f$$

Here,

f_c = center frequency

Δf = frequency deviation

M = Number of symbols/levels

Example of MFSK

- For a 4-FSK system, the center frequency is 250 kHz and the frequency deviation is 25 kHz. Determine (a) the number of bits per symbol, and (b) the individual carrier frequencies used.

$$\leadsto m = 4$$

$$F_c = 250 \text{ kHz}$$

$$F_d = 25 \text{ kHz}$$

$$\leadsto n = \log_2 m$$

$$= \log_2 4$$

$$= 2 \text{ bits/symbol}$$

$$f_i = f_c + (2i - M - 1)\Delta f$$

$$- F_1 = 250 + (2 - 4 - 1) 25 = 175 \text{ kHz}$$

$$F_2 = 250 + (4 - 4 - 1) 25 = 225 \text{ kHz}$$

$$F_3 = 250 + (6 - 4 - 1) 25 = 275 \text{ kHz}$$

$$F_4 = 250 + (8 - 4 - 1) 25 = 325 \text{ kHz}$$

Phase Shift Keying -

Outline

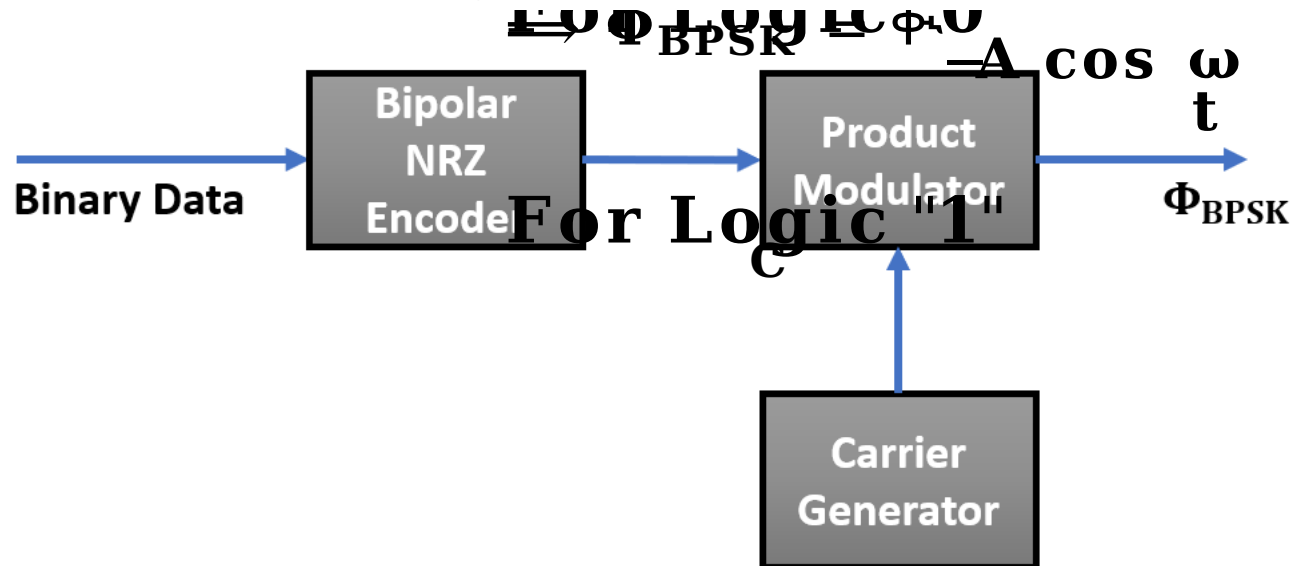
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- ☐ Waveforms of BPSK
- ☐ Bandwidth of PSK
- ☐ Multilevel PSK
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- ☐ Applications of PSK

Basics of PSK

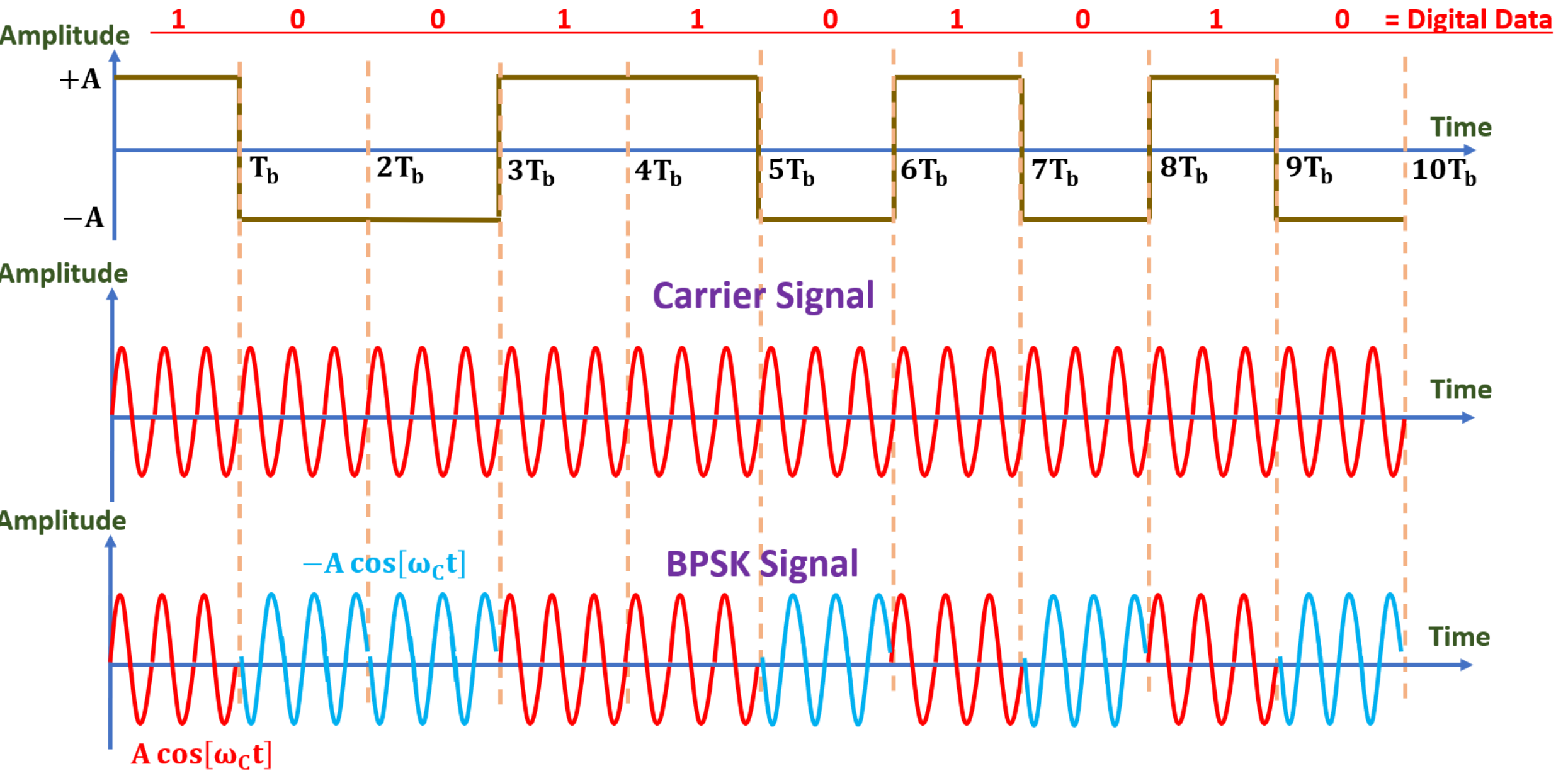
- ❑ PSK is a digital modulation technique that converts digital signals into Analog form by varying the carrier's Phase, enabling efficient transmission over a communication channel.
- ❑ In PSK, the Phase of the Carrier Signal varies according to the Digital Message Signal.
- ❑ Let's consider Binary PSK, the Binary message Signal $m(t)$, which is a

$$\Rightarrow \Phi_{\text{BPSK}} = \begin{cases} A \cos[\omega_c t] & \text{For Logic "1"} \\ -A \cos[\omega_c t] & \text{For Logic "0"} \end{cases}$$

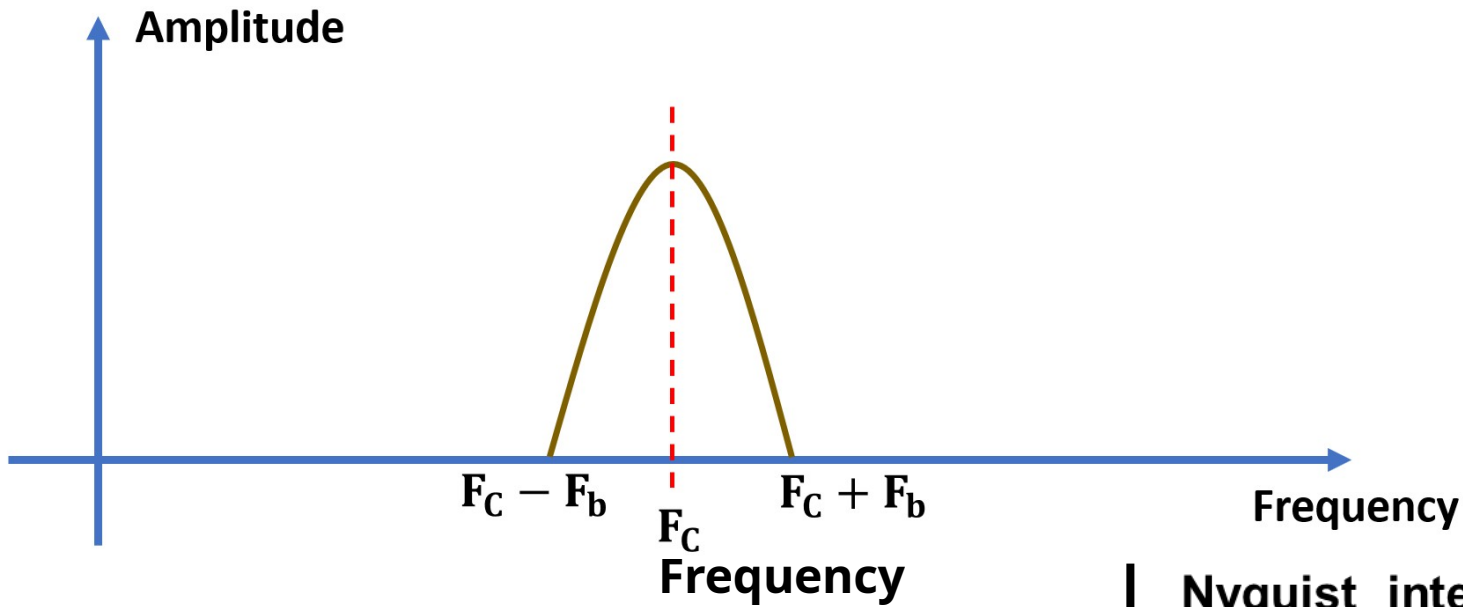
$\Rightarrow \phi = 0^\circ$
 $\Rightarrow \phi = 180^\circ$



Waveforms of PSK



Bandwidth of PSK



$$\Rightarrow BW_{PSK} = (1 + \alpha)r$$

- r is Baud Rate of Message Signal
- α is Roll off Factor depends on Filtering

$$BW_{min} = (1 + 0)r = r \text{ [as } \alpha = 0 \text{ during } BW_{min}]$$

Nyquist interval: the **minimum time gap between two consecutive symbols** so that they don't interfere with each other.

$$\text{Formula: } T_s = \frac{1}{2F_b}$$

[F_b = Highest baseband Frequency or baseband signal's BW]

$$\text{So, } F_b = \frac{1}{2T_s} = \frac{r}{2} \text{ [Here, } r \text{ is baud rate]}$$

BW of bandpass signal = 2 * baseband signal

$$\text{So, } BW_{min} = 2F_b = r \text{ [ignoring roll off factor]}$$

Multilevel PSK

~) For BPSK

$$- M = 2 = 2^1, \quad n = 1, \quad \phi = 0^\circ, 180^\circ$$

0	1

Digital data

~) For QPSK = 4PSK

$$- M = 4 = 2^2, \quad n = 2, \quad \phi = 0^\circ, 90^\circ, 180^\circ, 270^\circ$$

00	01	10	11

digital data

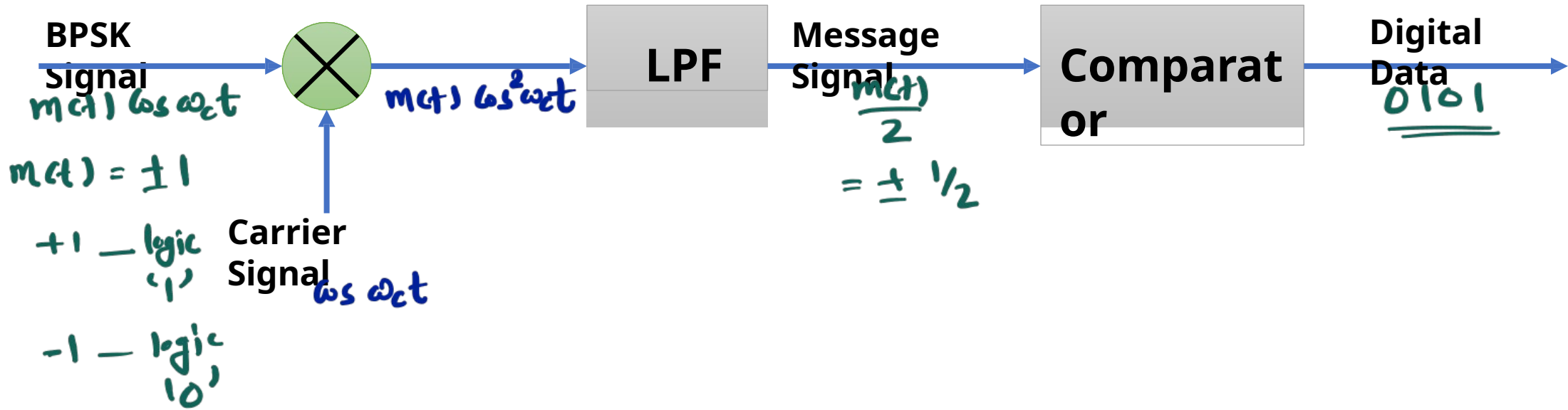
~) For 8-PSK

$$- M = 8 = 2^3, \quad n = 3, \quad \phi = 0^\circ, 45^\circ, 90^\circ, 135^\circ, 180^\circ, 225^\circ, 270^\circ, 315^\circ$$

000	001	010	011	100	101	110	111

digital data

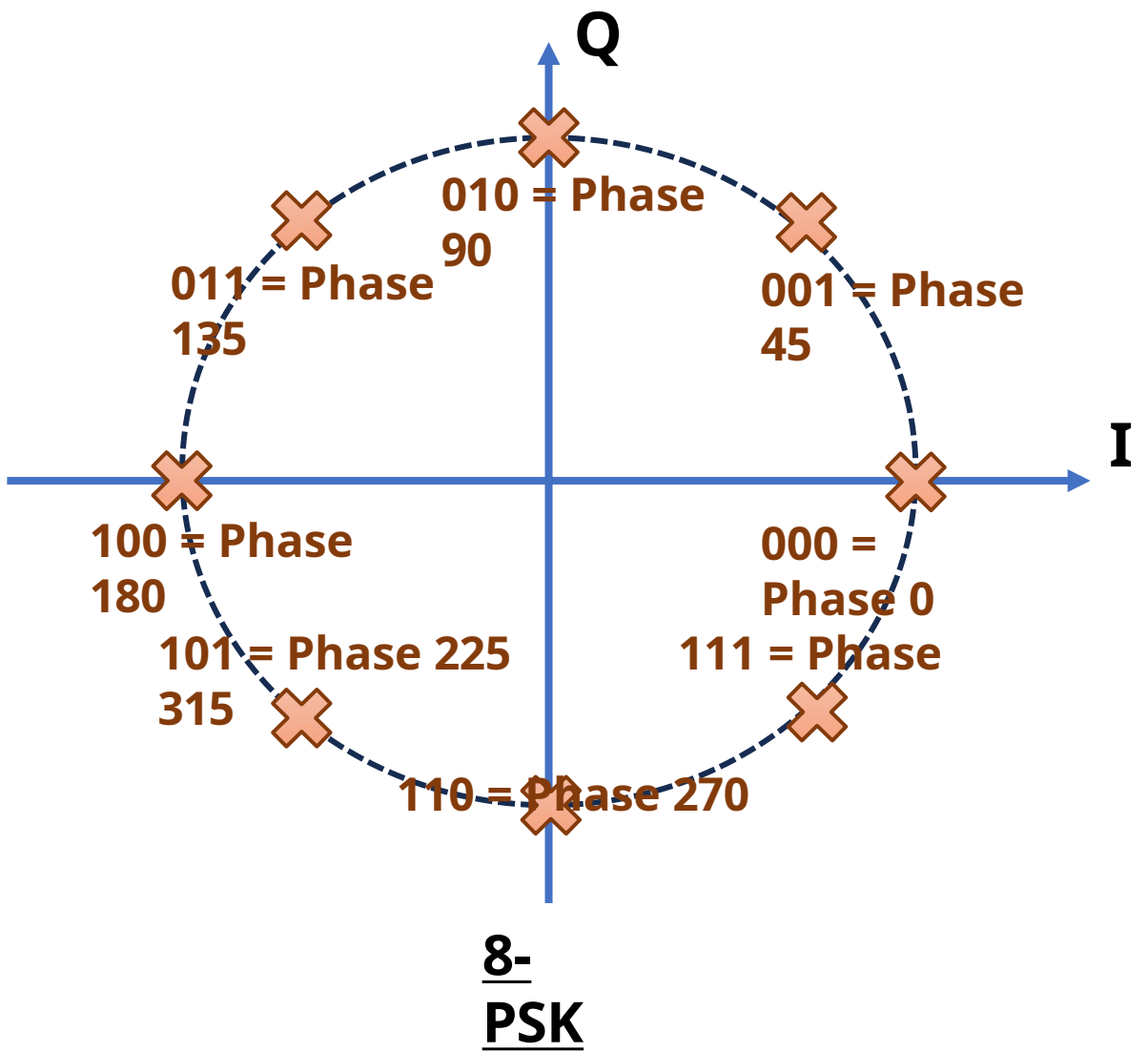
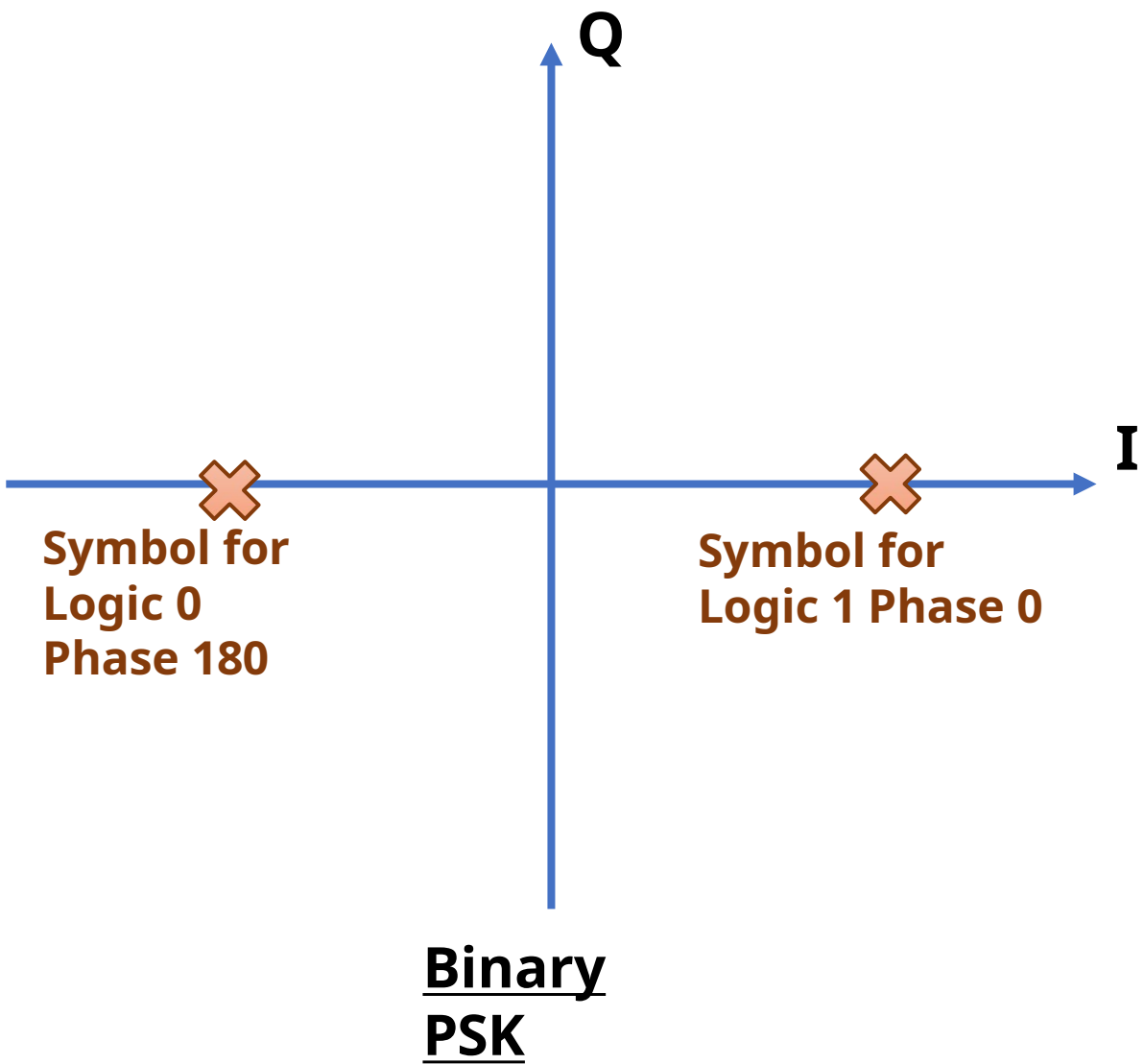
Demodulation of PSK



$$\begin{aligned}
 \Rightarrow \text{O/p Multiplier} &= m(t) \cos^2 \omega_c t \\
 &= m(t) \left(\frac{1 + \cos 2\omega_c t}{2} \right) \\
 &= \frac{m(t)}{2} + \frac{m(t)}{2} \cos (2\omega_c t)
 \end{aligned}$$

X
LPF eliminates

Constellation Diagram of PSK



Advantages of PSK

- ❑ Better Power Efficiency
- ❑ Better Bandwidth Efficiency
- ❑ Constant Envelope

Disadvantages of PSK

- ❑ Phase synchronisation is required.
- ❑ Sensitive to Phase Noise and Doppler.
- ❑ Higher Noise with Multipath Fading.
- ❑ Envelope detection is not possible.

Applications of PSK

- ❑ Wi-Fi {Uses BPSK and QPSK}
- ❑ Cellular {LTE/5G uses QPSK}
- ❑ Satellite {DVB-S2 uses 8-PSK}

Differential Phase Shift Keying

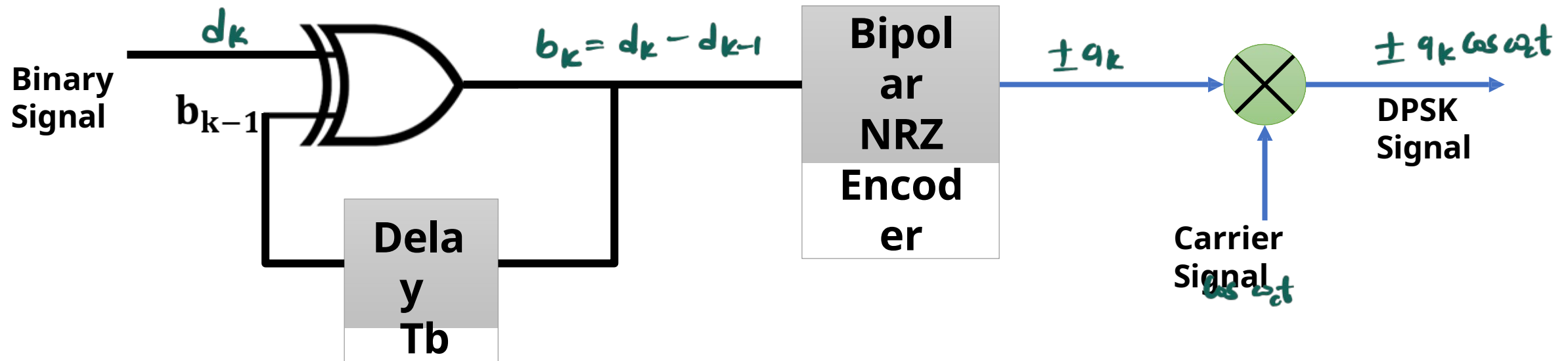
Outlines

- ❑ Basics of DPSK
- ❑ DPSK Transmitter
- ❑ DPSK Waveforms
- ❑ DPSK Receiver
- ❑ Advantages of DPSK
- ❑ Disadvantages of DPSK
- ❑ Applications of DPSK

Basics of DPSK

- ❑ DPSK (Differential Phase Shift Keying) is a type of Phase Shift Keying (PSK) modulation.
- ❑ Unlike standard PSK, DPSK encodes data based on the phase difference between successive bits instead of the absolute phase.
- ❑ Non-coherent detection of PSK is difficult.
- ❑ Non-coherent detection of DPSK reduces the cost of the circuits.

DPSK Transmitter



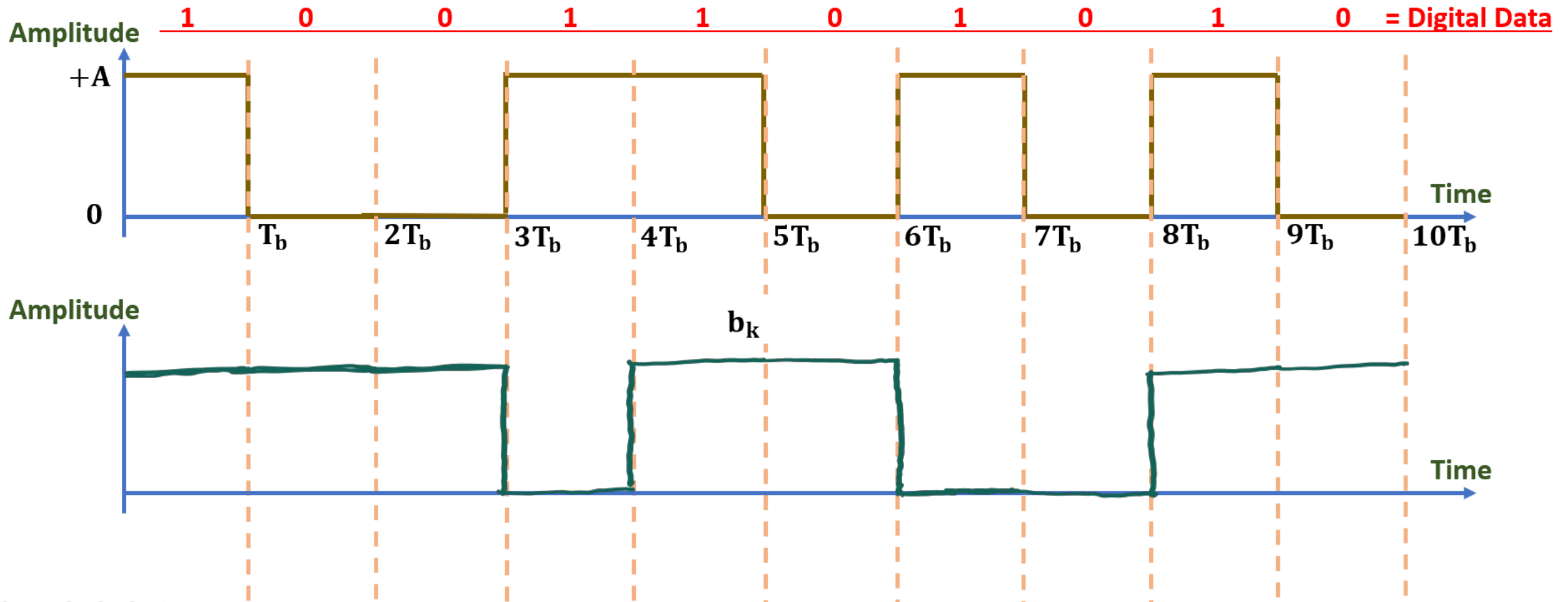
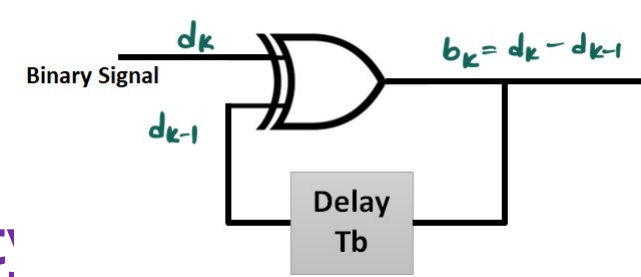
- ❑ DPSK Transmitter is similar to a PSK Transmitter.
- ❑ The binary signal is XORed with its previous bit before being applied to the PSK modulator.

XOR

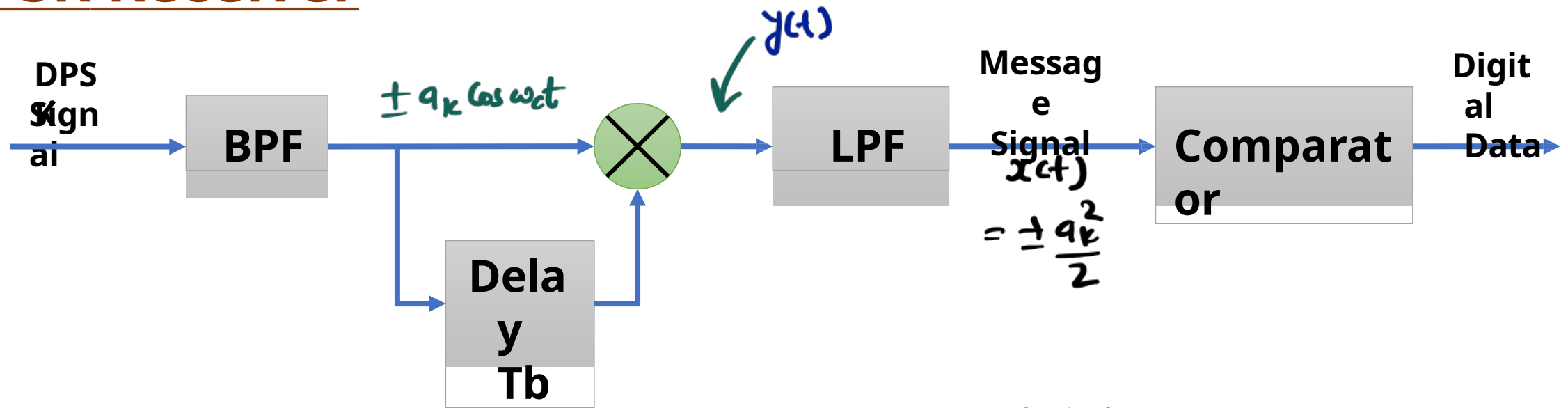
Input 1	Input 2	Output
0	0	0
0	1	1
1	0	1
1	1	0

DPSK waveforms

- ❑ First Input bit = First Output bit.
- ❑ If the next input bit is logic "1", then change the polarity of Output.
- ❑ If the next input bit is logic "0", then do not change the



DPSK Receiver



→ Case-I (Same polarity of DPSK and Delayed bit).

$$\Rightarrow y(t) = (a_k \cos \omega_c t) (a_k \cos \omega_c t)$$

$$= a_k^2 \cos^2 \omega_c t$$

$$= a_k^2 \left(1 + \frac{\cos 2\omega_c t}{2} \right)$$

$$= \frac{a_k^2}{2} + \frac{a_k^2 \cos(2\omega_c t)}{2}$$

X (LPF)

$$\xrightarrow{\text{LPF}} x(t) = \frac{a_k^2}{2}$$

Logic

'0'

2) Case-II (Opposite polarity of DPSK and Delayed bit).

$$\Rightarrow y(t) = (a_k \cos \omega_c t)(-a_k \cos \omega_c t)$$

$$= -a_k^2 \cos^2 \omega_c t$$

$$= -a_k^2 \left(\frac{1 + \cos 2\omega_c t}{2} \right)$$

$$= -\frac{a_k^2}{2} - \frac{a_k^2}{2} \cos(2\omega_c t)$$

⊗ LPF

↓ LPF

$$x(t) = -\frac{a_k^2}{2} \longrightarrow$$

Logic

'1'

Advantages of DPSK

- ☐ No Need for a Coherent Reference Carrier
- ☐ More robust to Phase Noise compared to Standard PSK
- ☐ Better Synchronisation Properties

Disadvantages of DPSK

- ☐ Poor Performance compared to Standard PSK
- ☐ Error Propagation {if one bit is wrongly detected, the next bit may also be affected}
- ☐ More susceptible to Noise and Interference

Applications of DPSK

- ☐ Wireless LAN and Bluetooth
- ☐ Optical Communication
- ☐ Satellite and Space Communication {Here, Coherent detection is difficult}
- ☐ Short Range RF Systems

BPSK {Singal Space Diagram}

1. Signal Representation

The two possible transmitted signals are

$$s_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t), \quad 0 \leq t \leq T_b$$

$$s_2(t) = -\sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t), \quad 0 \leq t \leq T_b$$

where

- E_b = energy per bit
- T_b = bit duration
- f_c = carrier frequency

The second signal is simply the **negative of the first**, meaning the phase difference is **180°**.

2. Orthonormal Basis Function

Choose the basis function

$$\phi_1(t) = \sqrt{\frac{2}{T_b}} \cos(2\pi f_c t), \quad 0 \leq t \leq T_b$$

It satisfies the orthonormality condition

$$\int_0^{T_b} \phi_1^2(t) dt = 1$$

Since both signals are multiples of the **same cosine waveform**, **only one basis function is required**.

Therefore

Signal space dimension = 1

3. Signal Vector Representation

The signal coordinate is

$$s_{i1} = \int_0^{T_b} s_i(t) \phi_1(t) dt$$

For $s_1(t)$

$$\begin{aligned} s_{11} &= \int_0^{T_b} \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t) \sqrt{\frac{2}{T_b}} \cos(2\pi f_c t) dt \\ &= \sqrt{E_b} \end{aligned}$$

So

$$s_1 = (\sqrt{E_b})$$

For $s_2(t)$

$$\begin{aligned} s_{21} &= \int_0^{T_b} -\sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t) \sqrt{\frac{2}{T_b}} \cos(2\pi f_c t) dt \\ &= -\sqrt{E_b} \end{aligned}$$

So

$$s_2 = (-\sqrt{E_b})$$

BPSK {Constellation Diagram or Signal Space

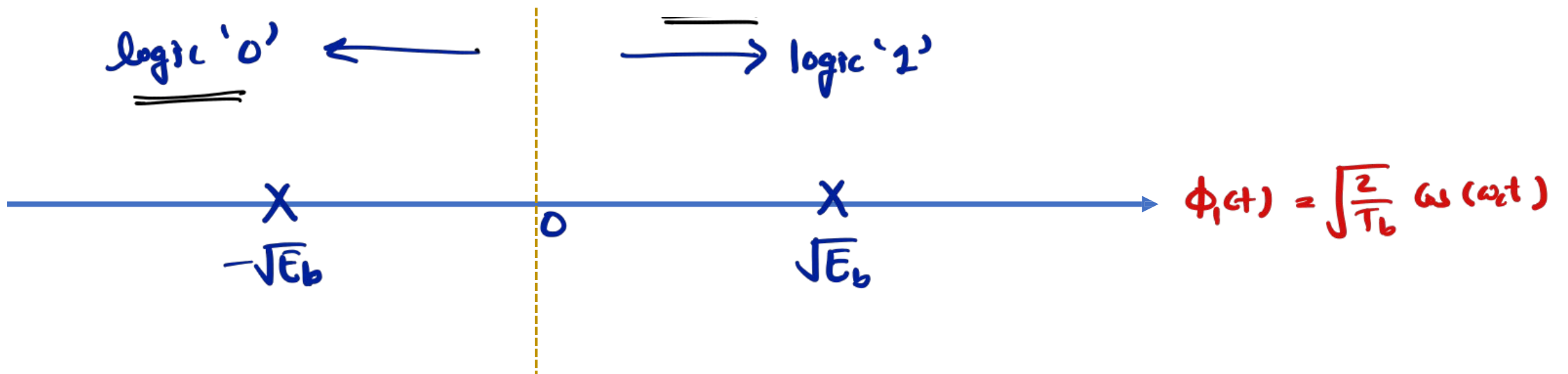
[

4. Signal Space Representation

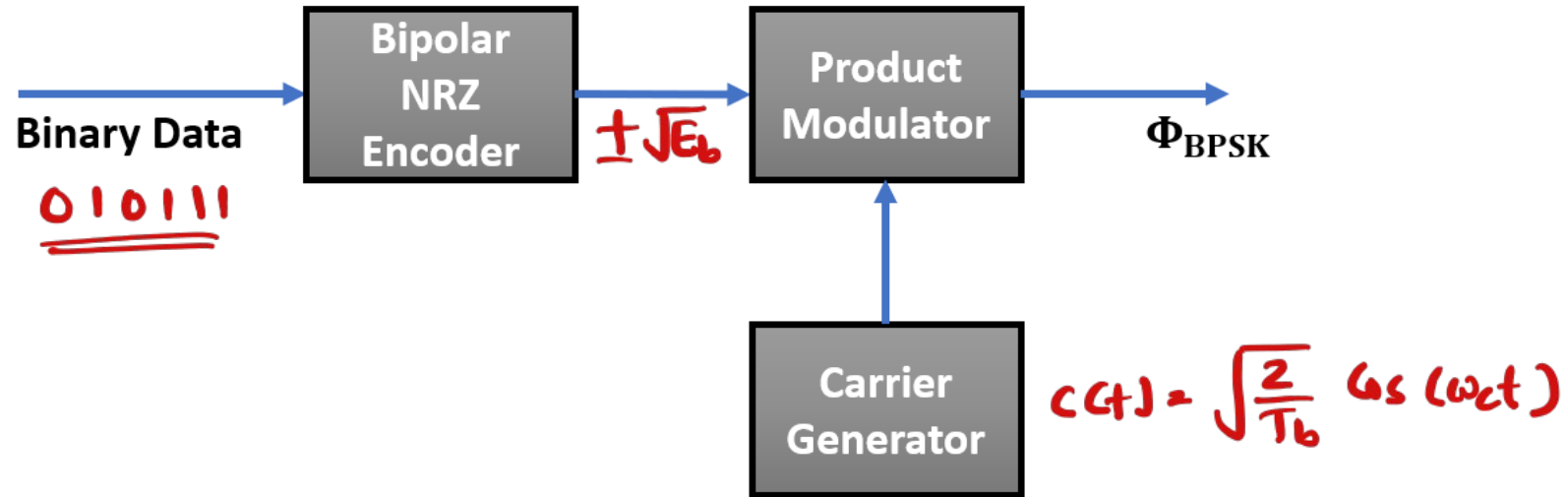
Since only one basis function exists, the signal space is 1-dimensional.

The constellation points are

$$+\sqrt{E_b} \quad \text{and} \quad -\sqrt{E_b}$$



BPSK Transmitter



BASK Vs BFSK Vs

Parameters	BASK	BFSK	BPSK
Modulation Characteristics	❖ Carrier Amplitude Varied	❖ Carrier Frequency Varied	❖ Carrier Phase Varied
Bandwidth (Min)	❖ $2F_b$ [F_b is the BW of baseband signal]	❖ $4F_b$	❖ $2F_b$
Noise Immunity	❖ Low	❖ High	❖ Moderate / High
Error Probability	❖ High	❖ Low	❖ Moderate / Low
Complexity	❖ Simple	❖ Moderate	❖ High
Bit rate	❖ Suitable for Low Bitrate	❖ Suitable for Moderate Bitrate	❖ Suitable for High Bitrate
Detection	❖ Envelope	❖ Envelope / Coherent	❖ Coherent
Power Efficiency	❖ Poor	❖ Moderate	❖ Excellent
Synchronization	❖ Low Requirement	❖ Moderate Requirement	❖ High Requirement
Phase Ambiguity	❖ None	❖ None	❖ Possible